

Road Sign Generator

GitHub: <https://github.com/katie34634/road-sign-generation>

Overview

This project uses a diffusion model to generate road signs. I trained the model using 22000 images from the German Traffic Sign Recognition Benchmark dataset, using only images that had a dimension of 32 pixels or higher. The intention was to generate novel road signs.

Generated Images at Epoch 400



Difficulties

For the first few attempts at training, the generated images were extremely deformed. I was using 10000 images, an image size of 128, a batch size of 256, a noise embedding size of 32, and training for 400 epochs. I thought that training for longer would improve the quality, so I

repeated training with 800 epochs, but the results were still very poor. I made several changes for the following reasons:

- Increasing the width of the unet — learns more details
- Increasing the dataset size to 22000
- Decreasing the image size to 64 — I think 128 was too large because most of the images in the dataset were not above 64 pixels
- Decreasing the batch size — more gradient updates for each epoch
- Increasing the noise embedding size to 128 — finer details

Once I made those changes the model was generating readable signs.

I noticed one sign that was not in the original dataset, which was a blank speed limit sign. I did not recognize any other images that were novel. I thought that if I trained long enough it would start to generate new signs, but the loss didn't appear to be decreasing any further.

I attempted to do DDIM inversion to get the noise that a real image would have started from, but the loss between the original and reconstructed image was really high so I pursued other types of latent space exploration.

Extra Criteria

By interpolating between noise vectors, I was able to generate some images that were clear(ish) and not in the original dataset. To do this, I took two random noise vectors and interpolated between them. Some of the interesting results:



Between blank speed limit (not in dataset) and 80 speed limit, there's one that kind of looks like π speed limit.



Backwards 2 somewhere between 30 speed limit and 70 speed limit.

I also looked into noise perturbation adding the same noise scaled by different values to some base noise vector. In some of the cases this actually made the base image appear clearer and brighter. Too much scaling of the perturbation made the noise exit the normal distribution and that showed gibberish, but you could kind of still see a shape.

